Week 6 Session A&B

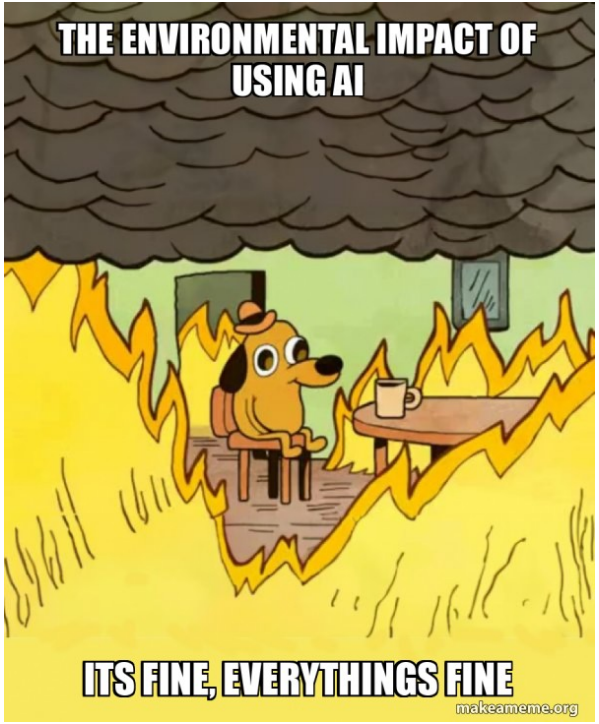
Thursday, Feb 19 2026

Overview

- Environmental Impact of AI
  - Multiple aspects of this topic
  - Data centers
- Week 6 Assignment

Environmental Impact of AI

- Activity:
  - Watch this video together
  - Let’s throw some keywords here~



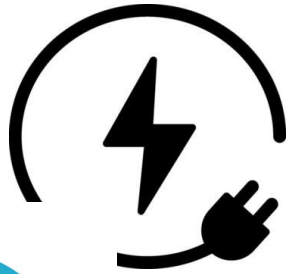
Environmental Impact of AI

- What's my role in this process?
  - interaction/inference
  - Consumer
  - Giving queries
  - Consent?

- What's the big tech's role in this process?
  - training!!!!!

Environmental Impact of AI I

- The proliferating data centres that house AI servers produce electronic waste. They are large consumers of water, which is becoming scarce in many places. They rely on critical minerals and rare elements, which are often mined unsustainably. And they use massive amounts of electricity, spurring the emission of planet-warming greenhouse gases.
  - United Nations Environment Programme "AI has an environmental problem. Here's what the world can do about that."

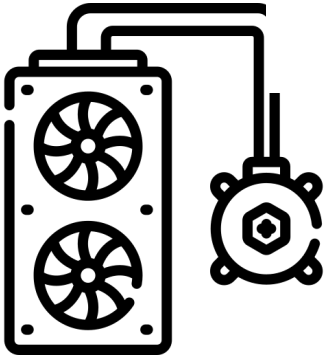


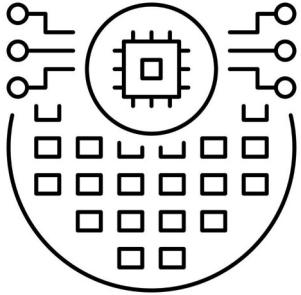
Environmental Impact of AI I

- **Water:**

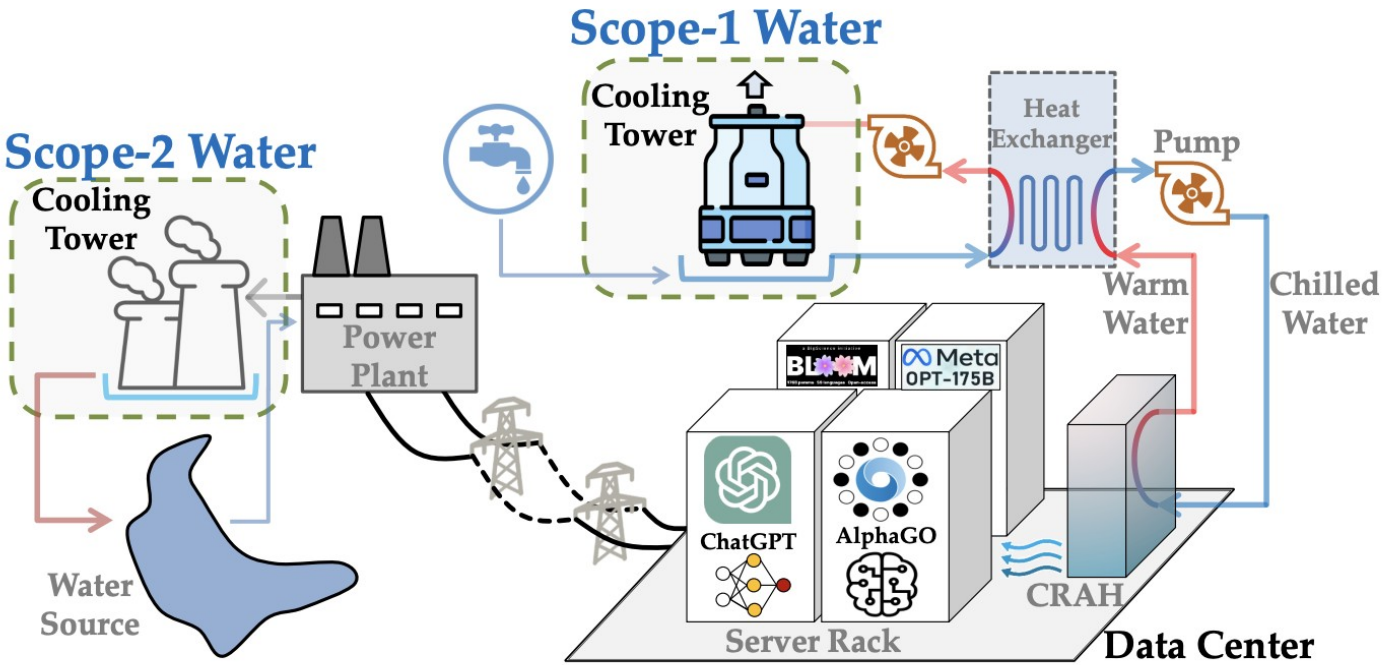
AI's water usage spans three scopes: on-site water for data center cooling (scope 1), off-site water for electricity generation (scope 2), and supply-chain water for server manufacturing (scope 3).

- Scope3: AI chip and server manufacturing uses a huge amount of water. Ultrapure water is needed for wafer fabrication and water is also needed for keeping semiconductor plants cool. The discharged water may contain toxic chemicals and/or hazardous wastes.





Environmental Impact of AI I



On-site scope-1 water usage for data center cooling (via cooling towers in the example), and off-site scope-2 water usage for electricity generation.

Environmental Impact of AI I

- ChatGPT:
  - What Sam Altman says?

  "People are often curious about how much energy a ChatGPT query uses; the average query uses about 0.34 watt-hours, about what an oven would use in a little over one second, or a high-efficiency lightbulb would use in a couple of minutes. It also uses about 0.000085 gallons of water; roughly one fifteenth of a teaspoon" (June 2025)
  - What other research shows?

Environmental Impact of AI I

- ChatGPT:
  - What other research shows?

Location	PUE	On-site WUE (L/kWh)	Off-site EWIF (L/kWh)	Water for Training (million L)			Water for Each Request (mL)			# of Requests for 500ml Water
				On-site Water	Off-site Water	Total Water	On-site Water	Off-site Water	Total Water	
U.S. Average	1.170	0.550	3.142	0.708	4.731	**5.439**	2.200	14.704	**16.904**	**29.6**
Arizona	1.180	1.630	4.959	2.098	7.531	**9.629**	6.520	23.406	**29.926**	**16.7**
Georgia*	1.120	0.060	2.309	0.077	3.328	**3.406**	0.240	10.345	**10.585**	**47.2**
Illinois	1.350	0.740	2.233	0.952	3.880	**4.833**	2.960	12.060	**15.020**	**33.3**
Iowa	1.160	0.140	3.104	0.180	4.634	**4.814**	0.560	14.403	**14.963**	**33.4**
Texas	1.280	0.250	1.287	0.322	2.120	**2.442**	1.000	6.590	**7.590**	**65.9**
Virginia	1.140	0.140	2.385	0.180	3.499	**3.679**	0.560	10.875	**11.435**	**43.7**
Washington	1.150	0.950	9.501	1.223	14.063	**15.285**	3.800	43.706	**47.506**	**10.5**
Wyoming	1.110	0.130	2.574	0.167	3.677	**3.845**	0.520	11.429	**11.949**	**41.8**
Australia*	1.120	0.012	4.259	0.015	6.138	**6.154**	0.048	19.078	**19.126**	**26.1**
Denmark*	1.160	0.010	3.180	0.013	4.747	**4.760**	0.040	14.754	**14.794**	**33.8**
Finland*	1.120	0.010	4.542	0.013	6.548	**6.561**	0.040	20.350	**20.390**	**24.5**
India*	1.430	0.000	3.445	0.000	6.340	**6.340**	0.000	19.704	**19.704**	**25.4**
Indonesia*	1.320	1.900	2.271	2.445	3.858	**6.304**	7.600	11.992	**19.592**	**25.5**
Ireland	1.190	0.020	1.476	0.026	2.261	**2.287**	0.080	7.027	**7.107**	**70.4**
Mexico*	1.120	0.056	5.300	0.072	7.639	**7.711**	0.224	23.742	**23.966**	**20.9**
Netherlands	1.140	0.060	3.445	0.077	5.054	**5.131**	0.240	15.708	**15.948**	**31.4**
Sweden	1.160	0.090	6.019	0.116	8.986	**9.101**	0.360	27.927	**28.287**	**17.7**

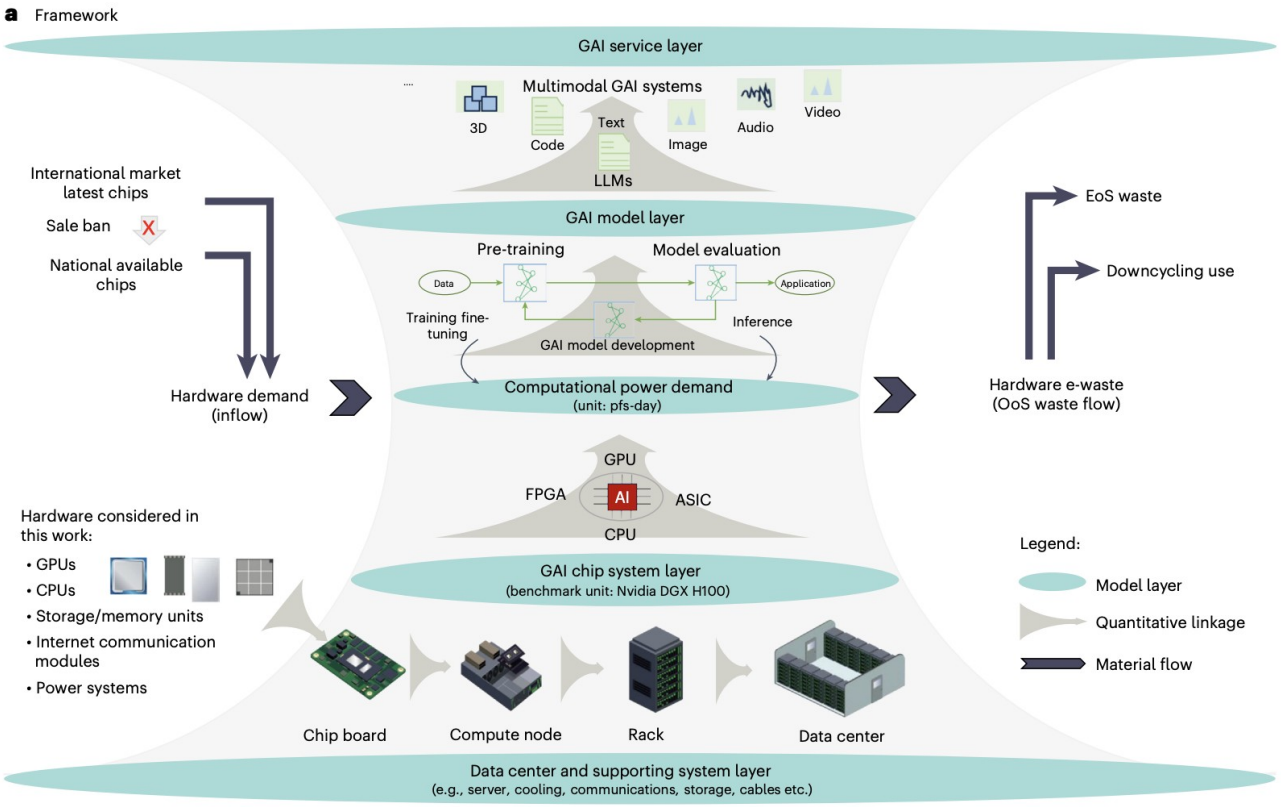
Environmental Impact of AI II

- Electricity:

Environmental Impact of AI III

- E-Waste:
  - Generative artificial intelligence (GAI) is driving a surge e-waste due to intensive computational infrastructure needs.
  - Our findings indicate that this e-waste stream could increase, potentially reaching a total accumulation of 1.2 –5.0 million tons during 2020–2030, under different future GAI development settings.
    (E-waste challenges of generative artificial intelligence)

Environmental Impact of AI III

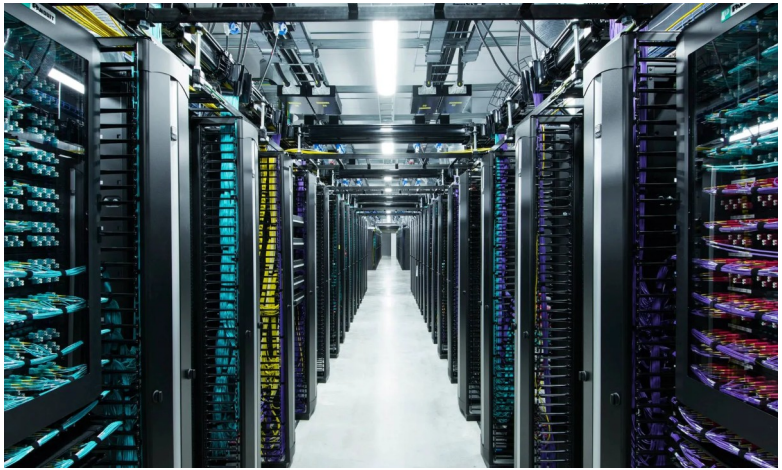


Environmental Impact of AI IV

- Mining:
  - A study by the non-governmental organization Justice et Paix (2019) notes that manufacturing a 2 kg computer involves the extraction of 800 kg of raw materials. Generally, manufacturing electronic devices requires 50 to 350 times of their final material weight. (2024 Digital economy report, 61)

Let’s just look at data centers

- ICER’s data center we visited
- Data center around the world
- Data center in the US
- Data center in Michigan
- Data center in Lansing (hopefully not)



Let’s just look at data centers

- ICER’s data center we visited
  - we don’t have a lot of facts from the internet. After this session, let’s think about what facts we would like to learn?
- Data center around the world:
  - From Statista
- From data center around the world, to data center in the US, to data center in Michigan, to data center in our area.
- Activity:
  - Data center map
  - Cloudscene

Let’s just look at data centers

Regions	Numbers	Notes

Local Data Centers (For next week)

- Saline Township
- Lansing

Week 6 Assignment